



جامعة النيلين

AL NEELAIN UNIVERSITY

كلية الدراسات العليا AI Master (2026)

Academic Context: COSC726 Agentic AI

Project Proposal:

Design of a Multi-Agent AI System for Training
Course Design and Student Evaluation



TO :

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1. Executive Summary & System Concept:

This project proposes the design of an intelligent, multi-agent platform designed to automate and elevate educational and training workflows. Traditionally, instructors face significant manual overhead in designing course outlines, allocating hours, formulating learning objectives, generating and grading assessments, analyzing student performance, and identifying knowledge gaps.

The proposed system transforms these fragmented manual processes into a streamlined **Agentic Workflow**. Specialized autonomous agents collaborate seamlessly, supported by deterministic software tools, structured Pydantic schemas, state management, human-in-the-loop validation gates, comprehensive execution tracing, and rigorous system evaluation.

2. Problem Statement & Proposed Solution :

Current Manual Challenges	Agentic AI Solution
High time consumption in structuring course hours, drafting syllabi, and writing content.	End-to-end automated lifecycle management from course planning to final reporting.
Difficulty in maintaining strict traceability between assessment questions and learning objectives.	Direct, hard-coded alignment between every assessment item and specific Learning Objectives (LOs).
Reliance on raw numerical scores (e.g., 60%) without deep diagnostic insights into strengths and weaknesses.	Detailed student proficiency profiling (e.g., identifying mastery in variables vs. weakness in loops/functions).
Absence of automated tracking for conceptual learning gaps across individuals and cohorts.	Autonomous recommendation engine generating targeted remedial actions and adaptive exercises.

3. System Architecture & Core Agents

The platform is structured around seven specialized agents coordinated by a central orchestrator:

Agent Name	Primary Responsibility	Key Output / Artifact
1. Course Designer Agent	Designs high-level course structure based on inputs.	Course Title, Modules, Hours, Prerequisites, Assessment Strategy.
2. Content Generator Agent	Generates detailed instructional content per module.	Lecture Topics, Code Examples, Exercises, Practical Tasks.
3. Assessment Agent	Produces objective-aligned assessments and tests.	MCQs, True/False, Coding Questions mapped strictly to LOs.
4. Student Evaluation Agent	Processes raw student submissions and answers.	Initial unit-level performance scoring and classification.
5. Learning Analyst Agent	Analyzes individual/class performance & finds gaps.	Critical Topic identification, Class Mastery percentages.
6. Recommendation Agent	Triggers pedagogical decisions based on thresholds.	Targeted remediation, extra exercises, or advanced tasks.
7. Orchestrator Agent	Manages workflow state, agent handoffs, and execution.	Coordinated agent pipeline execution and state persistence.

4. Technical Stack & Tool Integration

To ensure high reliability and eliminate LLM arithmetic/hallucination vulnerabilities, the system integrates robust programmatic tools:

- **Programming Language & Framework:** Python, built iteratively using custom ModelClient + Pydantic schemas before scaling to CrewAI or OpenAI Agents SDK.
- **Structured Outputs & State:** Pydantic models ensuring strict type-safe structured data (CoursePlan, TrainingState) across agent handoffs.
- **Database & Persistence:** SQLite database managing relational schemas (courses, modules, learning_objectives, questions, student_results, recommendations).
- **Deterministic Tools:** Grade Calculator, Course/Student DB interfaces, Question Bank reader, Report Generator, Statistics, and Notification utilities.

5. Engineering Guarantees & Reliability

- **Human-in-the-Loop (HITL):** Critical actions (such as modifying course structures or overriding failing thresholds) require explicit instructor approval via interactive gates.
- **Tracing & Auditability:** Comprehensive logging of agent inputs, tool outputs, confidence scores, and reasoning steps, enabling complete audit transparency.
- **System Evaluation:** Automated evaluation metrics measuring objective coverage, module consistency, classification accuracy, and tool-call correctness.

6. Implementation Milestones

The project will be developed incrementally across eight structured phases to ensure modular validation:

- **Phase 1: Core Agent & Course Design:** Implement Course Designer Agent with Pydantic CoursePlan schema.
- **Phase 2: Database & Grading Tools:** Establish SQLite database schemas and integrate deterministic grading tools.

- **Phase 3: Content Generation:** Build Content Generator Agent for lectures, examples, and exercises.
- **Phase 4: Assessment Generation:** Develop Assessment Agent mapping questions strictly to Learning Objectives.
- **Phase 5: Student Evaluation:** Enable student evaluation processing and score calculations.
- **Phase 6: Learning Analytics:** Activate Learning Analyst Agent to detect critical gaps and class performance metrics.
- **Phase 7: Recommendation Engine:** Implement Recommendation Agent for adaptive remedial and advanced tasks.
- **Phase 8: Multi-Agent Orchestration & UI:** Integrate all agents under Orchestrator workflow and build Streamlit Trainer Dashboard.